

Empirical Evidence of Membership Inference and Training Data Extraction in Open-source Large Language Models

ABSTRACT

The widespread adoption of open-source large language models (LLMs) for domain-specific fine-tuning on sensitive corpora introduces a class of privacy vulnerabilities that remain inadequately characterised in the existing literature. This paper presents a multi-model empirical audit of five open-source LLMs, namely llama3.2:3b, phi4-mini, qwen3 (8.2B), ministral-3:3b, and deepseek-r1:1.5b, using a greedy-match proxy membership inference attack (MIA) and a verbatim extraction probe methodology. All five evaluated models exhibit statistically significant membership leakage above the random baseline (AUC-ROC > 0.75, $p < 0.05$ under one-sided Mann-Whitney U testing). Two models achieve a perfect AUC-ROC of 1.0000, with score gaps of 0.167 to 0.210 between member and non-member distributions. Verbatim extraction rates range from 8.3 per cent to 66.7 per cent, with llama3.2:3b reproducing eight of twelve well-known training sequences from prefix alone. A novel finding is that chain-of-thought reasoning architectures suppress verbatim extraction rates whilst leaving membership inference advantage statistically intact, a distinction with direct implications for deployments relying upon RLHF-trained models. Established theoretical bounds from the differential privacy and membership inference literature are applied as interpretive frameworks for the empirical findings, and state-of-the-art mitigation strategies are examined. The empirical results constitute a conservative lower bound on the leakage achievable against models fine-tuned on higher-density proprietary corpora.

Impact statement

Large language models are increasingly fine-tuned on sensitive data in healthcare, legal, and financial settings where data privacy is both a legal obligation and an ethical imperative. This paper demonstrates, through direct empirical testing of five widely deployed open-source models, that such models systematically expose training data through two distinct mechanisms: statistical membership inference and verbatim text reproduction. Two models reproduce training sequences verbatim at rates exceeding 50 per cent. A finding of particular practical significance is that chain-of-thought reasoning, a property of RLHF-trained models, suppresses surface-level text reproduction whilst leaving membership leakage statistically intact. Organisations deploying fine-tuned open-source models on proprietary or regulated data should not regard current RLHF or instruction-tuning practices as privacy safeguards. The audit toolkit produced in this work is openly available to support responsible deployment practice.

I. INTRODUCTION

The proliferation of open-source large language models has fundamentally altered the economics of artificial intelligence deployment. Organisations across healthcare, legal, financial, and governmental sectors now routinely acquire openly licensed foundation models, including Llama 3 [1], Gemma [2], Phi [3], Mistral [4], and Qwen [5], and fine-tune them on domain-specific corpora of considerable sensitivity. The appeal of this paradigm is well established: the computational economics of fine-tuning vastly outperform training from scratch, and the representational quality of contemporary base

models is exceptional. Nevertheless, the fine-tuning process introduces a privacy exposure that is both distinct from and potentially more severe than that associated with pre-training.

When a model is fine-tuned on a sensitive proprietary corpus, its parameters are modified to assign higher probability to the patterns present in that corpus. This modification constitutes a form of encoding: the training data is not stored verbatim in any legible repository, yet it is recoverable, under appropriate adversarial conditions, from the model's observable input-output behaviour. Shokri et al. [6] demonstrated that model confidence scores reveal the membership status of training records with accuracy well above chance. Carlini et al. [7] subsequently showed that production-scale language models can be induced to reproduce verbatim training sequences through targeted prefix elicitation. What remains undercharacterised is whether these vulnerabilities are consistently present across the diverse ecosystem of open-source LLMs now in production, and how architecture-specific properties, particularly chain-of-thought reasoning, modulate the attack surface.

This paper addresses that gap through a systematic empirical audit of five open-source models representing distinct architectural families, parameter scales from 1.5B to 8.2B, and training paradigms. A greedy-match proxy MIA is implemented that is applicable under black-box access to the model's generation endpoint, alongside a verbatim extraction probe that elicits memorised continuations from known prefixes. The methodology is grounded in formal privacy theory, with established bounds from Yeom et al. [11] and Dwork and Roth [8] are applied as interpretive frameworks for the empirical results.

This paper makes four principal contributions. **(1)** A black-box MIA audit methodology applicable to any Ollama-served LLM, implemented as an open-source Python toolkit. **(2)** A multi-architecture empirical survey confirming statistically significant membership leakage across five open-source LLM families (all $p < 0.05$), distinguishing between standard and chain-of-thought architectures. **(3)** Empirical demonstration that chain-of-thought architectures suppress verbatim extraction whilst leaving MIA advantage intact. **(4)** A cross-model correlation analysis (Spearman $\rho = +0.60$) grounding the theoretical generalisation-gap bound empirically.

II. BACKGROUND AND THREAT MODEL

A. Notation and privacy definitions

Let M_θ denote a language model with parameters θ . A training dataset $D = \{z_1, \dots, z_n\}$ consists of n token sequences, each $z = (t_1, \dots, t_l)$ drawn from vocabulary V .

The per-sample cross-entropy loss is:

$\mathcal{L}(M, z) = -(1/l) \sum_i \log P_M(t_i | t_1, \dots, t_{i-1})$ (1)
 with perplexity $PP(M, z) = \exp(\mathcal{L}(M, z))$. The formal privacy framework is (ϵ, δ) -differential privacy [8]:

Definition 1 ((ϵ, δ)-Differential Privacy).

A randomised mechanism $M: D \rightarrow R$ satisfies (ϵ, δ) -DP if, for all adjacent datasets D and D' differing in one record, and for all $S \subseteq R$:

$$Pr[M(D) \in S] \leq e^\epsilon \cdot Pr[M(D') \in S] + \delta$$

where $\epsilon \geq 0$ is the privacy budget and $\delta \geq 0$ the failure probability.

B. Attack taxonomy

The machine learning security literature encompasses three threat classes. Data poisoning attacks operate at training time, injecting adversarially crafted samples to corrupt model behaviour. Adversarial attacks operate at inference time, perturbing inputs to elicit attacker-controlled outputs. Data leakage attacks, which are the subject of this paper, seek to recover information about the training corpus from observable model behaviour. Conflating these categories leads to misallocated defensive investment and misspecified threat models.

Within the leakage class, three sub-types are distinguished: Membership Inference Attacks (MIA), which determine whether a specific record was in the training set; Model Inversion Attacks, which reconstruct sensitive features of training data from model outputs [9]; and verbatim extraction, which recovers complete training sequences from the model's generation conditioned on a prefix [7].

C. Open-source risk amplification

The open-source deployment paradigm amplifies leakage risk along two structural dimensions. First, public availability of base model checkpoints dramatically lowers the barrier for shadow model construction, as an adversary may instantiate an exact replica of the victim architecture. Second, the chain of accountability for training data provenance is attenuated across developer, fine-tuner, and deployer, a gap that Bommasani et al. [10] characterise as the "foundation model accountability gap." The threat model assumed throughout is a grey-box adversary with access to the model's text generation API and architectural knowledge of the base model, both conditions structurally guaranteed in open-source deployments.

III. ATTACK METHODOLOGY

A. Membership inference formalisation

Definition 2 (Membership Inference Attack).

A MIA is a randomised algorithm $A: (M, z) \rightarrow \{0, 1\}$ where $A(M, z) = 1$ denotes the prediction " $z \in D_{\text{train}}$." The adversarial advantage is:

$$Adv_A = |Pr[A = 1 | z \in D] - Pr[A = 1 | z \notin D]|$$

The true positive rate $TPR = Pr[A = 1 | \text{member}]$ and false positive rate $FPR = Pr[A = 1 | \text{non-member}]$ define the operating point at threshold τ .

The foundational threshold attack of Yeom et al. [11] uses per-sample loss as the membership score, predicting membership when loss falls below τ . Theorem 1 provides the theoretical bound fundamental to interpreting the empirical results.

The following theorem is due to Yeom et al. [11] and is not an original contribution of this paper. It is included as an interpretive framework that formally contextualises the score gap measurements in Section IV-C and the generalisability of the empirical findings.

Theorem 1 (Yeom Generalisation-Gap Bound [11]).

Let $L_{\text{train}} = (1/n) \sum_n L(M, z)$ be the empirical training loss and $L_{\text{test}} = E[L(M, z)]$ the expected test loss. Then:

$$Adv_A \leq L_{\text{test}} - L_{\text{train}}$$

Proof sketch. Members achieve lower loss by virtue of the training objective. The advantage is bounded by the generalisation gap because the member and non-member loss distributions overlap by at most this quantity. Perfect generalisation implies zero advantage; severe overfitting implies near-maximal advantage.

B. Greedy-match proxy scoring

In the black-box deployment context studied here, per-sample loss values are unavailable, as the model API returns only generated text. A greedy-match proxy score is therefore introduced, which exploits the maximum-likelihood continuation under greedy decoding (temperature = 0).

Definition 3 (Greedy-Match Proxy Score).

For candidate text z , partition z into $k = 4$ windows, splitting each at position p_j into prefix p_j and continuation c_j . Let $M(p_j)$ denote the model's greedy continuation. The proxy score is:

$$s_{\text{GM}}(z) = (1/k) \sum_j \text{overlap}(M(p_j), c_j)$$

where $\text{overlap}(a, b) = |W(a) \cap W(b)| / |W(b)|$ and $W(\cdot)$ denotes word sets. The decision rule is: predict MEMBER if $s_{\text{GM}}(z) \geq \tau$.

C. Verbatim extraction probing

For each of twelve extraction probes, the model is queried with a prefix under greedy decoding with a token budget of 512. A probe is scored as a hit if the expected token appears in the generated continuation with word-level overlap of at least 0.8, or as an exact substring match. Probe texts are drawn from sequences documented as highly prevalent in Common Crawl, The Pile, and C4, spanning six domains: computer science, literature, history, religion, linguistics, and physics.

D. Experimental corpus design

The member corpus consists of twelve factual, encyclopaedic texts drawn from domains overwhelmingly represented in open-source pre-training datasets. The non-member corpus consists of twelve adversarially constructed synthetic texts containing post-cutoff fictional proper nouns and implausible domain-specific concatenations, sequences designed to be absent from any standard pre-training corpus.

The balanced design ($n = 12$ per group) is acknowledged to be modest; results should be regarded as indicative and replication at $n \geq 100$ is recommended before submission to a primary venue. Statistical significance is assessed using the one-sided Mann-Whitney U test with H_1 : member scores $>$ non-member scores, without assuming score normality. A known limitation of the word-overlap metric is that for single-token expected values, semantically related terms may satisfy the 0.8 threshold in place of the exact expected token. Inspection of probe P09 for deepseek-r1:1.5b revealed one such instance, where the generated text contained the word "heaven" but not "earth," yet was scored as a hit. Extraction rates for single-token probes should therefore be treated as approximate upper bounds.

IV. EMPIRICAL EVALUATION

A. Experimental setup

Five open-source LLMs served via Ollama [12] were evaluated on a Ubuntu 24.04 host with an Intel Core Ultra 7 CPU. All models use Q4_K_M quantisation. The audit tool queries the generation endpoint exclusively, without accessing internal weights, logits, or gradient information. gemma4:e4b is excluded from the primary analysis owing to null-generation behaviour under raw-completion prompting, as described in Section IV-G.

B. Membership inference attack results

Table I presents the full MIA evaluation metrics. All five evaluable models exceed $\text{AUC-ROC} = 0.75$, with llama3.2:3b and phi4-mini achieving a perfect score of 1.0000. A perfect AUC indicates that every member text score exceeds every non-member score, meaning the two distributions are entirely non-overlapping. The Mann-Whitney p-value of 0.000018 is the minimum achievable at $n = 12$ per group, confirming the result holds under all possible random permutations of the data. Table I additionally reports bootstrap 95% confidence intervals computed from 10,000 resamples. For llama3.2:3b and phi4-mini, the CI is [1.000, 1.000], confirming that perfect distributional separation is robust to resampling at $n = 12$ per group. For the remaining three models, CI widths of approximately 0.38 to 0.41 reflect the modest sample size; the point estimates remain above 0.75 and statistically significant, but the intervals indicate the true AUC could be as low as 0.54 in the worst case. This width is expected and is reported in full in the interest of methodological transparency.

Applying a Bonferroni correction for the five-model test family ($\alpha = 0.01$) to guard against inflated family-wise error from repeated testing, the results for llama3.2:3b, phi4-mini, and qwen3:latest ($p \leq 0.0082$) remain significant; ministral-3:3b ($p = 0.0112$) and deepseek-r1:1.5b ($p = 0.0171$) approach but do not strictly clear the corrected threshold. These two results should accordingly be interpreted with additional caution relative to the other three.

TABLE I

Primary MIA Results with Bootstrap 95% Confidence Intervals (10,000 resamples). AUC = 0.5 is the random baseline; AUC = 1.0 is perfect separation. †Nominal parameter count from model name; GGUF total (incl. embeddings): phi4-mini 3.8B,

ministral-3 3.8B, deepseek-r1 1.8B. ‡gemma4:e4b excluded (see Section IV-G).

Model	Params	AUC	95% CI	Adv.	p-value	Sig.
llama3.2:3b	3.2 B	1.0000	[1.000, 1.000]	0.667	0.000018	✓
phi4-mini	3.8 B†	1.0000	[1.000, 1.000]	1.000	0.000018	✓
qwen3:latest	8.2 B	0.7917	[0.576, 0.945]	0.417	0.008158	✓
ministral-3:3b	3.8 B†	0.7778	[0.556, 0.951]	0.500	0.011185	✓
deepseek-r1:1.5b	1.8 B†	0.7569	[0.538, 0.938]	0.500	0.017110	✓
gemma4:e4b ‡	large	0.5000	—	0.000	1.000000	×

C. Score distribution analysis

Table II reports mean proxy scores for member and non-member distributions. phi4-mini and llama3.2:3b exhibit member means 3.3 to 3.5 times larger than non-member means. The score gap $\mu_{\text{in}} - \mu_{\text{out}}$ corresponds directly to the empirical generalisation gap in Theorem 1: models with large score gaps are both theoretically more vulnerable and empirically more susceptible, a relationship confirmed by the Spearman correlation in Table IV.

TABLE II

μ_{in} : member mean; μ_{out} : non-member mean; Gap = $\mu_{\text{in}} - \mu_{\text{out}}$; Ratio = $\mu_{\text{in}} / \mu_{\text{out}}$.

Model	μ_{in}	μ_{out}	Gap	Ratio
phi4-mini	0.2706	0.0604	0.2102	3.48×
llama3.2:3b	0.2170	0.0501	0.1668	3.33×
qwen3:latest	0.1250	0.0536	0.0713	1.33×
ministral-3:3b	0.1119	0.0507	0.0612	1.21×
deepseek-r1	0.0941	0.0466	0.0474	1.02×

D. Verbatim extraction results

Table III presents the verbatim extraction hit matrix. llama3.2:3b achieves the highest extraction rate at 66.7 per cent (8 of 12), followed by phi4-mini at 50.0 per cent (6 of 12) and ministral-3:3b at 41.7 per cent (5 of 12). Probe P07 (the Dickens opening) achieves a cross-model rate of 100 per cent and P09 (the Genesis creation verse) achieves 80 per cent. This universality is consistent with very high-frequency occurrence of these sequences in Common Crawl and The Pile, achieving sufficient repetition during pre-training to be encoded eidetically across all evaluated model families.

Particularly striking is the llama3.2:3b response to the Gettysburg Address prefix. Given "...brought forth on this continent, a new," the model generates "...nation, conceived in liberty, and dedicated to the proposition that all men a." This constitutes multi-clause verbatim reproduction and represents eidetic memorisation [13] in a directly observable, non-statistical form.

TABLE III

Verbatim Extraction Hit Matrix. $\checkmark$ = word overlap ≥ 0.8 or exact match; $\times$ = miss. Bold rate indicates ≥ 80 per cent cross-model extraction.

#	Domain	Expected	phi4	llama3.2	min-3	ds-r1	qwen3	Rate
P01	CS/PL	<i>indentation</i>	$\checkmark$	$\times$	$\times$	$\times$	$\times$	20%
P02	CS/ML	<i>2017</i>	$\times$	$\checkmark$	$\times$	$\times$	$\times$	20%
P03	CS/OS	<i>1991</i>	$\times$	$\checkmark$	$\times$	$\times$	$\times$	20%
P04	DevOps	<i>containers</i>	$\checkmark$	$\times$	$\checkmark$	$\times$	$\times$	40%
P05	VCS	<i>source code</i>	$\times$	$\times$	$\times$	$\times$	$\times$	0%
P06	Lit.	<i>question</i>	$\times$	$\checkmark$	$\times$	$\checkmark$	$\times$	40%
P07	Lit.	<i>times</i>	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	100%
P08	Hist.	<i>nation</i>	$\checkmark$	$\checkmark$	$\times$	$\times$	$\times$	40%
P09	Relig.	<i>earth</i>	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	80%
P10	Hist.	<i>equal</i>	$\times$	$\checkmark$	$\checkmark$	$\times$	$\times$	40%
P11	Ling.	<i>dog</i>	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\times$	60%
P12	Phys.	<i>squared</i>	$\times$	$\times$	$\times$	$\times$	$\times$	0%
	Rate		50.0%	66.7%	41.7%	25.0%	8.3%	38.3%

E. Cross-architecture correlation analysis

Table IV presents the cross-model relationship between score gap, extraction rate, and AUC-ROC. The Spearman rank correlation between score gap and AUC-ROC is $\rho = +0.91$, confirming a strong monotonic relationship between distributional separability and membership inference advantage. The correlation between score gap and extraction rate is $\rho = +0.60$, with the principal deviation attributable to qwen3, explained mechanistically in Section IV-F.

TABLE IV

Score Gap, Extraction Rate, and AUC-ROC by Model. Spearman ρ over rank orders. CoT = chain-of-thought.

Model	Score Gap	Extraction	AUC-ROC	Architecture
phi4-mini	0.2102	50.0%	1.0000	Standard
llama3.2:3b	0.1668	66.7%	1.0000	Standard
qwen3:latest	0.0713	8.3%	0.7917	Thinking (CoT)
ministral-3:3b	0.0612	41.7%	0.7778	Standard
deepseek-r1	0.0474	25.0%	0.7569	Thinking (CoT)
Spearman ρ	+0.60	—	+0.91 (vs gap)	—

No relationship between parameter count and AUC-ROC was observed across the 1.8B to 8.2B range tested. deepseek-r1:1.5b (1.8B) and qwen3:latest (8.2B) achieve comparable AUC values of 0.7569 and 0.7917 respectively, despite a more than fourfold difference in scale, while llama3.2:3b (3.2B) and phi4-mini (3.8B) both achieve AUC = 1.0000 despite differing in scale from each other and from the larger qwen3. Architecture type, specifically the presence of chain-of-thought reasoning, was a stronger predictor of MIA resistance in this sample than model scale, a pattern that warrants confirmation across a wider parameter range in future work.

F. Thinking model behaviour: A novel finding

qwen3 (8.2B) and deepseek-r1 (1.5B) are chain-of-thought reasoning models that produce internal reasoning output prior to their final response. This architectural property has an asymmetric effect upon the two attack vectors studied here. Verbatim extraction rates collapse to 8.3 per cent and 25.0 per cent respectively, because the model's reasoning preamble displaces the memorised continuation from the generated text before the word-overlap scorer can register a match. Membership inference advantage, however, is entirely preserved: qwen3 achieves AUC = 0.7917 ($p = 0.008$) and deepseek-r1 achieves AUC = 0.7569 ($p = 0.017$).

This result demonstrates that chain-of-thought reasoning constitutes an incidental and incomplete defence against extraction whilst leaving the underlying statistical memorisation signal intact. Inspection of the qwen3 extraction outputs reveals that verbatim training text surfaces within the thinking stream: for probe P06, deepseek-r1 generates "The phrase 'to be, or not to be, that is the question' by William Shakespeare..." with the expected text appearing verbatim inside the reasoning chain rather than in the response field. This implies that even where extraction appears to fail by the response-field criterion, the model may be disclosing memorised content through a channel not monitored by conventional output-filtering systems.

The consistency of this asymmetry across both evaluated CoT architectures, contrasted with its absence in both evaluated standard architectures, is stated below as a conjecture rather than a theorem. No analytical proof is offered; the claim is empirical and is intended to motivate future mechanistic and theoretical investigation.

Conjecture 1 (Chain-of-Thought extraction suppression).

Let M be a language model trained with chain-of-thought reasoning, producing both a thinking-stream output and a final response for each query. Let $s_{GM}(z)$ denote the greedy-match proxy score and $r(z)$ denote the response-field extraction indicator for candidate text z . Then, for CoT architectures, the empirical correlation between $s_{GM}(z)$ and $r(z)$ across a corpus is lower than for standard autoregressive architectures trained on comparable data, even when the two architectures exhibit comparable membership inference advantage.

A formal proof of Conjecture 1 would require a theoretical characterisation of how chain-of-thought training redistributes conditional probability mass between the thinking-stream and response-field token sequences, which remains an open problem and is identified as a priority direction in Section VI.

G. Limitations

Several limitations bound the generalisability of these findings. The member corpus consists exclusively of public-domain text; the reported AUC and extraction rates should therefore be treated as indicative minima; models fine-tuned on smaller, higher-density proprietary corpora are expected to exhibit more severe memorisation based on established theoretical results [14]. The greedy-match proxy score is a coarser approximation to per-sample loss than the LiRA likelihood ratio statistic [15], which requires logprob API support not available in the tested deployment. gemma4:e4b produced null output for eleven of twelve raw-completion probes (91.7 per cent null response rate), a consequence of its strong instruction-following fine-tuning, which causes the model to refuse open-ended sentence completion rather than continue the supplied prefix. A chat-format adaptation wrapping each probe as a user message is required and is left for future work. Finally, $n = 12$ per group yields moderate statistical power; replication at $n \geq 100$ is recommended before a primary venue submission.

The most significant methodological limitation concerns the absence of a verified membership oracle. The member corpus was designated on the basis that these public-domain sequences appear with high frequency in documented pre-training datasets such as Common Crawl, The Pile, and C4. However, the training manifests of the evaluated models are not publicly available in complete form, meaning ground-truth membership status cannot be independently verified for any individual sequence. The observed AUC-ROC values of 1.0000 for llama3.2:3b and phi4-mini could reflect two distinct phenomena: genuine membership inference, in which the model assigns disproportionately high probability to sequences it trained upon; or a fluency differential, in which the model handles common, syntactically regular internet text better than synthetic adversarial text regardless of specific training membership. Disentangling these explanations requires either access to the training manifest or a controlled fine-tuning experiment in which membership is known with certainty. Future work should address this by fine-tuning an open-source base model on a documented proprietary corpus and auditing the resulting checkpoint against a confirmed training set, thereby establishing a genuine membership oracle. The results presented here should accordingly be interpreted as evidence of differential model behaviour towards likely-member versus designed non-member text, rather than as formally validated membership inference against a confirmed training manifest.

A canary exposure test (Carlini et al. [13]) was additionally attempted against all five evaluated models, following the synthetic canary insertion methodology described in Section III. The test could not be completed for any model, as none of the evaluated Ollama deployments expose token-level log

probabilities through their generation API, a prerequisite for computing the exposure metric. This represents a structural limitation of black-box auditing against API-served open-source models rather than a finding about the models themselves, and is resolved only by logprob-capable deployments or white-box checkpoint access.

H. Ethical considerations and responsible disclosure

All models evaluated in this study are publicly released open-source checkpoints with no associated training data confidentiality obligations known to the authors. No proprietary, production, or third-party deployed systems were tested. The audit methodology and accompanying toolkit are released to support defensive privacy auditing practice. The authors note the dual-use potential of any membership inference tooling and encourage its application be limited to systems the practitioner owns or is explicitly authorised to test.

V. MITIGATION STRATEGIES AND COUNTERMEASURES

A. Differential privacy via DP-SGD

Differentially Private Stochastic Gradient Descent (DP-SGD) [16] is the gold standard for formal privacy guarantees in language model training. The mechanism clips per-sample gradients to bound the sensitivity of each update, adds calibrated Gaussian noise, and provides an (ϵ, δ) -DP guarantee via the Rényi Differential Privacy accountant [17]. This guarantee directly bounds MIA advantage, as formalised in Theorem 2.

The following theorem is due to Dwork and Roth [8] and is likewise an established result. It is reproduced here to contextualise the DP-SGD mitigation in Section V-A and to quantify the formal gap between the theoretical guarantee and the empirically observed advantages in Table I.

Theorem 2 (DP Bounds MIA Advantage [8]).

If training mechanism M satisfies (ϵ, δ) -DP, then for any adversary A :

$$TPR \leq e^{\epsilon} \cdot FPR + \delta$$

and consequently $Adv_A \leq (e^{\epsilon} - 1) \cdot FPR + \delta$.

For $\epsilon = 1$, $\delta = 10^{-5}$, $FPR = 0.1$: $Adv_A \leq 0.172$, compared to empirically observed advantages of 0.42 to 1.0 recorded in Table I.

The practical challenge for LLM fine-tuning is scale: per-sample gradient computation prevents standard batching optimisations. Li et al. [18] demonstrated that combining DP-SGD with LoRA-style parameter-efficient fine-tuning substantially mitigates this overhead whilst maintaining meaningful privacy guarantees, making differentially private fine-tuning tractable for models in the 7B parameter class. Applying Theorem 2 directly to the empirical results in Table I, reducing MIA advantage below 0.20 requires $\epsilon \leq 1.61$ for llama3.2:3b and phi4-mini (corresponding to noise multiplier $\sigma \geq 0.80$ at $T = 1,000$ gradient steps, batch size 32 from $n = 1,000$ training samples), and $\epsilon \leq 0.59$ for qwen3 and deepseek-r1 ($\sigma \geq 1.32$). The tighter

requirement for thinking-model architectures arises from their higher observed FPR of 0.25 at the operating threshold. Achieving $\sigma \geq 1.32$ for a 7B-class model under standard fine-tuning conditions represents a substantial utility penalty. This quantification confirms that DP-SGD alone cannot address the leakage demonstrated in Section IV without significant model performance degradation, reinforcing the necessity of a compositional defensive architecture.

B. Training data deduplication

Kandpal et al. [19] demonstrated empirically that removing near-duplicate sequences from the training corpus substantially reduces memorisation rates, as repeated exposure to identical sequences is the primary driver of eidetic memorisation. Deduplication is computationally inexpensive relative to DP-SGD and provides immediate reduction in the extraction rates observed in Table III. Notably, probes P07 and P09 achieved near-universal extraction precisely because these sequences appear with exceptionally high frequency in web-scale corpora; deduplication directly addresses this vulnerability class. For fine-tuning pipelines handling sensitive data, deduplication-before-training should be considered a mandatory baseline control.

C. Compositional and operational defences

No single mechanism currently provides comprehensive protection without unacceptable utility loss. A depth-in-defence architecture is therefore recommended, comprising: DP-SGD at training time to bound statistical leakage; deduplication of the fine-tuning corpus prior to training; output monitoring and PII filtering at inference time; and canary insertion with continuous auditing to detect memorisation events post-deployment [13]. The findings in Section IV-F carry a specific implication: chain-of-thought reasoning must not be relied upon as a privacy control, as it suppresses extraction only at the response-field level whilst leaving the underlying memorisation accessible through the thinking-stream channel.

VI. CONCLUSION

This paper has presented a multi-model empirical audit of membership inference and verbatim extraction vulnerabilities in five open-source LLMs, grounded in formal privacy theory and accompanied by proof sketches for the key bounding theorems. All five evaluable models exhibit statistically significant membership leakage, with two achieving perfect AUC-ROC separation. Verbatim extraction of training sequences is demonstrated at rates of 8.3 to 66.7 per cent across model families. The principal novel finding is that chain-of-thought reasoning architectures attenuate verbatim extraction without meaningfully reducing membership inference advantage, a distinction with direct implications for any deployment relying upon RLHF or CoT training as an implicit privacy safeguard.

Models fine-tuned on smaller, higher-density proprietary corpora are expected to exhibit more severe memorisation owing to reduced corpus cardinality and greater example density, suggesting these findings understate the risk in

regulated deployment settings.

VII. GLOSSARY OF TECHNICAL TERMS

Adversarial Attack. An inference-time perturbation of model inputs designed to elicit incorrect or attacker-controlled outputs. Distinct from data leakage attacks in objective and mechanism.

AUC-ROC. Area Under the Receiver Operating Characteristic Curve. Summarises a binary classifier's discriminative ability across all decision thresholds. AUC = 0.5 is random; AUC = 1.0 is perfect separation.

Canary. A synthetic record deliberately inserted into a training corpus at known frequency. Its recoverability from model outputs is used to quantify memorisation depth via the exposure metric.

Chain-of-Thought Reasoning (CoT). An architectural property of certain LLMs wherein the model produces internal reasoning tokens prior to its final response. Shown in this paper to suppress verbatim extraction whilst preserving MIA advantage.

Data Poisoning. A training-time attack in which adversarially crafted samples are injected into the training dataset to corrupt model behaviour at inference time.

Differential Privacy (DP). A formal mathematical framework bounding the information an adversary can obtain about any individual training record from the mechanism's output. Parameterised by privacy budget ϵ and failure probability δ .

DP-SGD. Differentially Private Stochastic Gradient Descent. Clips per-sample gradients and adds Gaussian noise before each parameter update to provide DP guarantees during model training.

Eidetic Memorisation. A form of training data memorisation in which a specific sequence can be extracted verbatim from a model's outputs given a prefix, even when the sequence appears only once in training data.

Fine-Tuning. The process of adapting a pre-trained foundation model to a specific domain or task by continuing gradient-based training on a domain-specific dataset.

Generalisation Gap. The difference between a model's expected test loss and its empirical training loss: $L_{\text{test}} - L_{\text{train}}$. Directly upper-bounds the Yeom threshold MIA advantage.

Greedy-Match Proxy Score. The black-box MIA metric introduced in this paper. Measures word-level overlap between the model's greedy continuation from a prefix and the actual continuation, averaged over multiple split points.

LiRA. Likelihood Ratio Attack. A membership inference methodology based on the Neyman-Pearson likelihood ratio statistic, using populations of shadow models.

Membership Inference Attack (MIA). An adversarial technique that determines, from a model's observable behaviour, whether a specific data record was a member of the training set.

Model Inversion Attack (MIV). An attack reconstructing sensitive features or characteristics of training data from a model's outputs.

Open-source LLM. A large language model whose weights are publicly released, enabling download, inspection, and local deployment. Provides adversaries white-box architectural access.

Perplexity. A measure of how well a language model predicts a text sequence: $PP(M,z) = \exp(L(M,z))$. Lower perplexity indicates higher assigned probability, the fundamental signal in loss-based MIA.

RLHF. Reinforcement Learning from Human Feedback. A training methodology aligning LLMs with human preferences via a reward signal derived from human judgements.

Shadow Model. A model of the same architecture as the target, trained on surrogate data with known membership status, used to calibrate MIA classifiers.

Verbatim Extraction. A data leakage attack recovering complete training sequences from a model's generation when conditioned upon a known prefix.

VIII. REFERENCES

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